Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Some concepts that have been covered that would help me in this project would be truthiness, loops, modifying/ creating arrays, conditionals, while loops, if statements, nested if statements. I want to improve my logic. I know how to do everything, but I need to know WHEN to do things. Additionally, I want to get better with DRAFTER.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easiest would most likely be retrieving data from the arduino in “bytes”, then converting into actual numbers and adding into the arrays. The harder ones would be to get a set count of measurements from the ultrasonic sensor, because it receives thousands of data points per second, but I want the user to choose how many data points they want, and the arduino should only give that many. That would be hard, another thing that would be hard would be using NumPY as I am a bit unfamiliar, as well as putting an image of the graph on Drafter, and just using Drafter in general.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Learning NUMPY and MatPlotLib, as well as implementing my final results on Drafter. The first thing that should be done is to actually receive the data from the arduino, and make sure I am accurately receiving it, and being able to choose the number of data points I want.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

I am unsure of what distractions might interfere with my work. While working I will need the Drafter Documentation by my side, as well as my Arduino Kit, all wired up for the Ultrasonic Sensor. I have no questions.

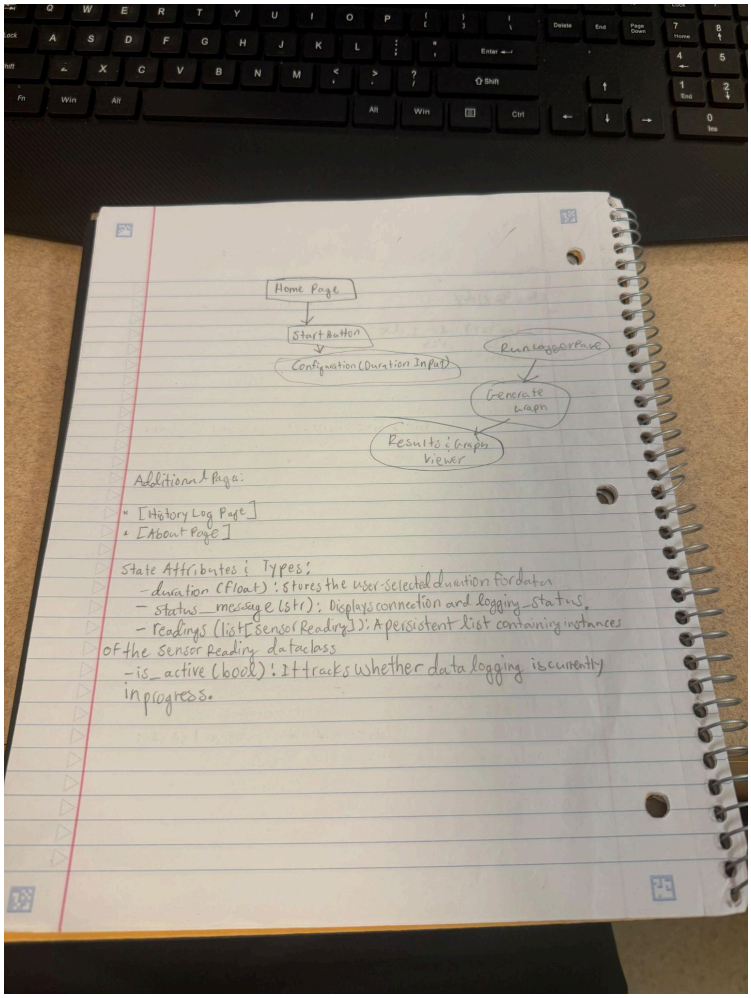
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	A web application built with Drafter that interfaces with Arduino Ultra-sonic sensor via PySerial to collect distance telemetry over a specified duration, perform linear regression using NumPY, and visualize distance vs time graph using MatPlotLib.pyplot

State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Duration: float Status_message: str Readings: list[SensorReading] Is_active: bool
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Class Sensor_Reading: Attributes: - Timestamp: float - Distance: float
Routes (minimum 7) The names of your route functions	A home route, a config (configuration route), collect (Data collection) route, results (graph viewer) route, history (log history) route, about route (about info, what the website does), reset route (reset state)
Pages (minimum 5) A brief description of the pages in your site	1. Home Page 2. Configuration Page 3. Collection & Progress Page 4. Results & Graph Page 5. History / Log Page
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these	1. TextBox: input duration in seconds 2. CheckBOX: toggle regression trend line visibility 3. SelectBOX: Choose COM Port for arduino connection

input types, see Drafter Documentation.	
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. Validate user input duration 2. Verify if serial data is non-empty for float conversion 3. Verify if serial data is even present, else use pre-setup arrays 4. Check if len(readings) >= 2 before rendering plot (you need two points for a line)
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	For loop / while loop iterating over readings (list of sensor readings instances), For reading in readings: Total_distance += reading.distance

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM for this.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.